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DECLARATION OF OWN WORK

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Unit 6: [16 marks: Engineering Science]

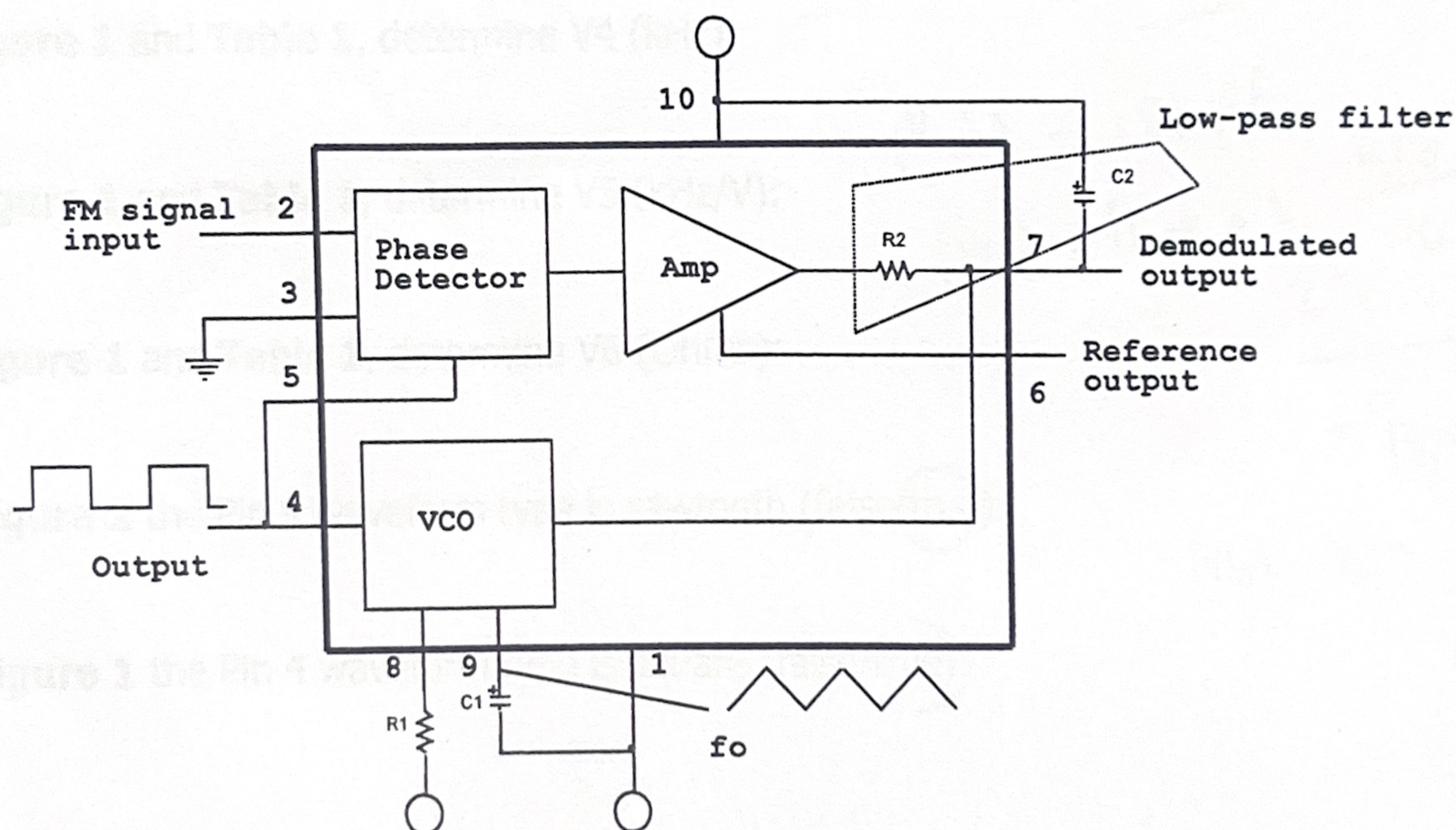


Figure 1: Unit 6 Q

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Table 1: Unit 6 Q

Parameter	UsedValues	Value	Units
Free-running frequency (fo)	Case 1: R1 = 2.15e+03 Ohm, C1 = 5.00e-10 F	V1	kHz
Calculated C1 for R1 and fo_{target}	Case 2: R1 = 6.00e+03 Ohm, fo_{target} = 2.40e+05 Hz	V2	pF
E24 capacitor for C1	Case 3: Calculated C1 from Case 2	V3	pF
PLL Lock Range	Case 4: fo from Case 1, V_{CC} = 20.0 V	V4	kHz
Conversion Gain (K)	Case 5: f1 = 95 kHz, f2 = 100 kHz, V1 = 1.5 V, V2 = 2.0 V	V5	kHz/V
E24 resistor for R1	Case 6: R1 from Case 1	V6	Ohm

From **Figure 1** and **Table 1**, determine V1 (kHz):
2

From **Figure 1** and **Table 1**, determine V2 (pF):
2

From **Figure 1** and **Table 1**, determine V3 (pF):
2

From **Figure 1** and **Table 1**, determine V4 (kHz):
2

From **Figure 1** and **Table 1**, determine V5 (kHz/V):
2

From **Figure 1** and **Table 1**, determine V6 (Ohms):
2

From **Figure 1** the Pin 9 waveform type is sawtooth (false/true):
2

From **Figure 1** the Pin 4 waveform type is square (false/true):
2

From **Table 2**, determine V6 (dimensionless):
2

Unit 5: [20 marks: Engineering Science]

From **Table 2**, determine V9 (dimensionless):
1

From **Table 2**, determine V10 (kHz):
2

$$(V2) = f_o = \frac{1}{4R_1C_1} = \frac{1}{4(2.15 \times 10^3)(5.00 \times 10^{-10})} = 279069.77$$

$$(V2) = C_1 = \frac{1}{4R_1f_o} = \frac{1}{4(6.00 \times 10^3)(279.07 \times 10^3)} = 179.17 \text{ pF}$$

$$(V3) = 181 \text{ pF}$$

$$(V4) = f_{lock} = \pm \frac{Bf_o}{V_{CC}} = \pm \frac{279.07 \times 10^3}{20} = 13.95 \text{ kHz/V}$$

$$(V6) K = \frac{\Delta f_o}{\Delta V_{CC}} = \frac{100 - 95}{2.0 - 1.5} = 10$$

[16]

$$(V5) \quad V_s/V_p = k \times \sqrt{\frac{L_s}{L_p}} = 0,015 \times \sqrt{\frac{75,00 \times 10^{-6}}{150 \times 10^{-6}}} = 0,011$$

$$= 0,011$$

$$= 10,61 \text{ m}$$

$$V = A_{vp} \times \frac{V_s}{V_p} \times \frac{V_o}{V_s}$$

$$1 \times 10^3 \times 10,61 \times 10^{-3}$$

$$\times 283,71 = 1,75 \text{ M}$$

$$\frac{V_o}{V_s} = \frac{2\pi f L_s}{R_s \times (75 \times 10^3)} = \frac{2\pi (1,838 \times 10^6)}{3} = 283,71$$

Table 2: Unit 5 Q

Given values: $L = L_p = 150.00 \mu\text{H}$, $CC = 4.00 \text{ pF}$, $C_p = 46.00 \text{ pF}$, $C = CC + C_p$, $RC = 110.00 \text{ k}\Omega$, $h_{fe} = 100$, $h_{ie} = 1.20 \text{ k}\Omega$, $k = 0.015$, $L_s = 75.00 \mu\text{H}$, $R_W = 3 \Omega$, $R_s = 3 \Omega$, $R_L = 7.50 \text{ k}\Omega$

Parameter	Value
fr	V1
ZP	V2
RL_prime	V3
AVP	V4
Vs/vp	V5
AV	V6
QP	V7
QL	V8
Q	V9
B (Hz)	V10

From **Table 2**, determine V1 (MHz)

2

$$(V1) \quad f_o = \frac{1}{2\pi \sqrt{L_p (C_p + CC)}} = 1837762,985 = 1,838 \text{ MHz}$$

From **Table 2**, determine V2 (Mega-ohms)

2

From **Table 2**, determine V3 (kilo-ohms)

2

$$(V2) \quad Z_p = \frac{L_p}{C_p + CC(R_W)} = \frac{150 \times 10^{-6}}{(4,00 \times 10^{-12} + (46,00 \times 10^{-12}) \times 3)} = 1000000 \text{ } 1 \text{ MHz}$$

From **Table 2**, determine V4 (dimensionless)(kilo)

2

From **Table 2**, determine V5 (dimensionless)(milli)

2

From **Table 2**, determine V6 (dimensionless)

2

$$(V3) \quad R_{L\text{-prime}} = Z_p \parallel R_c \parallel R_L = \left(\frac{1}{1 \times 10^6} + \frac{1}{110,00 \times 10^3} + \frac{1}{7,50 \times 10^3} \right)^{-1} = 6972,32 \Omega = 6,972 \text{ k}\Omega$$

From **Table 2**, determine V7 (dimensionless)

2

From **Table 2**, determine V8 (dimensionless)

2

$$(V4) \quad A_{VP} = \frac{h_{fe} \times R'_{L}}{h_{ie}} = \frac{100 \times 6,972 \times 10^3}{1,2 \times 10^3} = 581 = 0,581$$

From **Table 2**, determine V9 (dimensionless)

2

From **Table 2**, determine V10 (kHz)

2

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[20]

Unit 4: [28 marks: Engineering Science]

Table 3: Unit 4 Q

Given values for a Butterworth high-pass filter: fc = 4.40 kHz, R1 = 12.00 kΩ, R4 = 15.00 kΩ	
-20 dB per decade stage	
Parameter (units)	Value
fc1 (kHz)	V1 ✓
C1 (nF)	V2 ✓
E24 of C1 (nF)	V3 ✓
R2 (kΩ)	V4 .
E24 of R2 (kΩ)	V5
-40 dB per decade stage	
Parameter (units)	Value
fc2 (kHz)	V6
C3 (nF)	V7
E24 of C3 (nF)	V8
C2 (nF)	V9
E24 of C2 (nF)	V10
R3 (kΩ)	V11
E24 of R3 (kΩ) (	V12
R5 (kΩ)	V13
E24 of R5 (kΩ)	V14

$$\frac{R_1}{1} = \frac{R_2}{2}$$
$$2 = 2R_1$$
$$= 2(12 \times 10^3)$$
$$R_2 = 24 \text{ k}\Omega$$
$$R_2 = R_3$$

(V11) $R_3 = R_2$
 $= 24 \text{ k}\Omega$

(V10) $= 3,3$

(V13) R_5
 $= \frac{\sqrt{2}}{(2\pi \times 3,52 \times 10^3)}$
 $\times 3,76 \times 10^{-9}$
 $= 17 \text{ k}\Omega$

(V14) $= 1,8$

From Table 3, determine V1 (kHz)
2

(V1) $f_{c1} = 0,65 f_c$
 $= 0,65 (4,40 \times 10^3)$
 $= 2860 = 2,86 \text{ k}\Omega$

From Table 3, determine V2 (nF)
2

(V2) $C_1 = \frac{1}{2\pi R_1 f_{c1}}$
 $= \frac{1}{2\pi (12,00 \times 10^3) (2,86 \times 10^3)}$
 $= 4,637 \text{ nF}$
 $= 4,7 \text{ nF}$

From Table 3, determine V3 (nF)
2

From Table 3, determine V4 (kilo-ohms)
2

(V4) $R_2 = R_1$
 $= 12,00 \text{ k}\Omega$

From Table 3, determine V5 (kilo-ohms)
2

From Table 3, determine V6 (kHz)
2

(V5) $1,2$

From Table 3, determine V7 (nF)
2

(V6) $f_{c2} = 0,8 f_c$
 $= 0,8 (4,40 \times 10^3)$
 $= 3520 = 3,52 \text{ kHz}$

From Table 3, determine V8 (nF)
2

(V10) $= 3,9 \text{ nF}$
 $C_2 = C_3 = 2(1,88 \times 10^{-9})$
 $= 3,76 \text{ nF}$

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$C_3 =$
 $C_2 = 2C_3$
(V7) $C_3 = \frac{1}{2\pi R_3 f_{c2}}$
 $= 1,48 \text{ nF}$

(V8) $2,2$

From **Table 3**, determine V9 (nF)

From **Table 3**, determine V10 (nF)

From **Table 3**, determine V11 (kilo-ohms)

From **Table 3**, determine V12 (kilo-ohms)

From **Table 3**, determine V13 (kilo-ohms)

From **Table 3**, determine V14 (kilo-ohms)

[28]

Unit 3: [16 marks: Engineering Science]

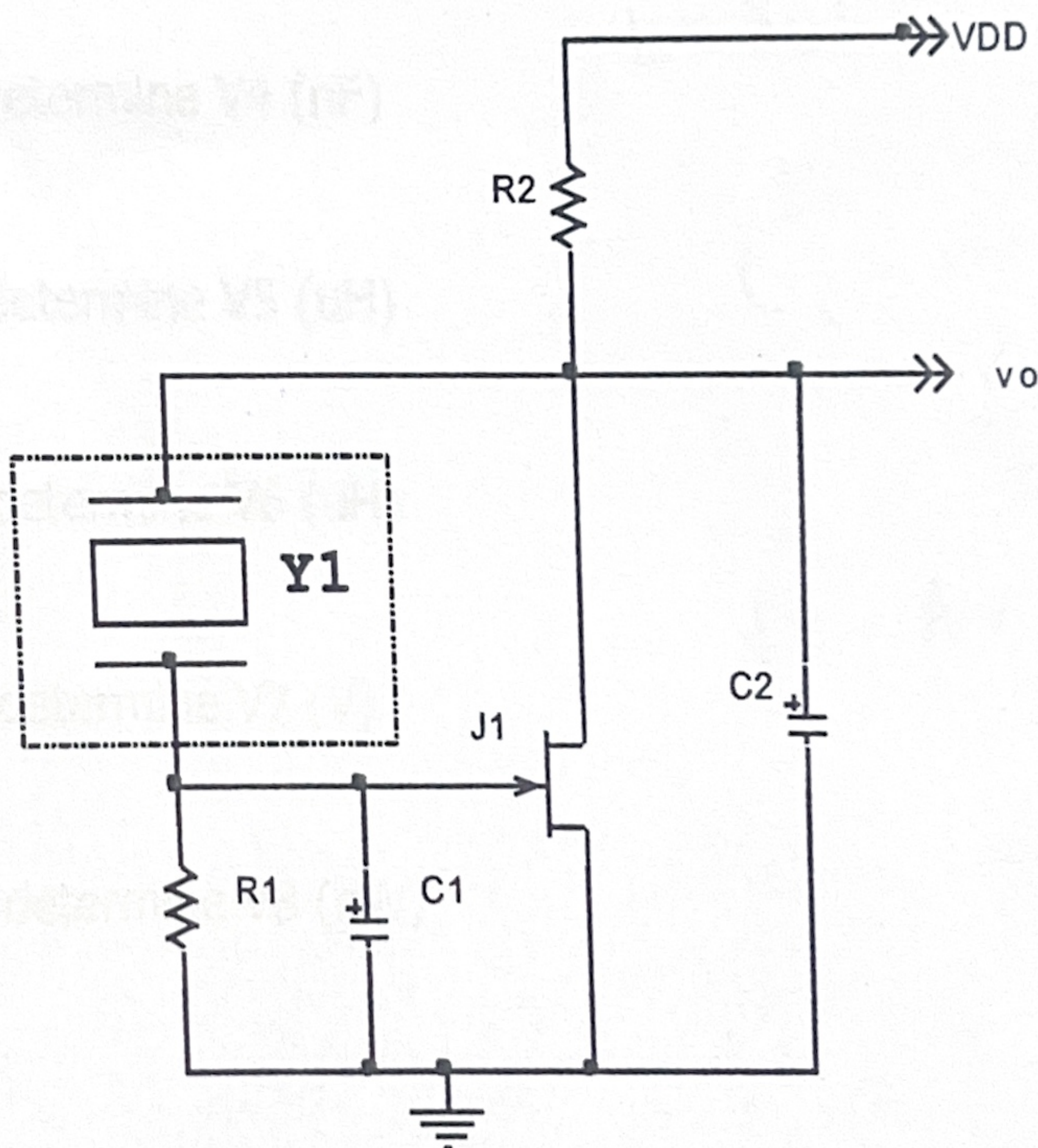


Figure 2: Unit 3 Q

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Table 4: Unit 3 Q

Given values: $f_s = 2 \text{ MHz}$, $R_s = 750 \text{ ohms}$, $(C_2/C_1) = 10$, $C_T = 100 \text{ pF}$, $X_L = X_{CT}$, $R_1 = 1.2 \text{ Mega-ohms}$, $R_2 = 15 \text{ kilo-ohms}$, $i_p = 8.23 \text{ uA}$	
Parameter (units)	Value
C1 (pF)	V1
C2 (nF)	V2
E24 of C1 (pF)	V3
E24 of C2 (nF)	V4
L (uH)	V5
E24 of L (uH)	V6
VDD (V)	V7
PD (nW)	V8

From **Figure 2** and **Table 4**, determine V1 (pF)
2

$$(V1) \quad C_1 = \frac{1}{2\pi f R_2}$$

$$= \frac{1}{2\pi (2 \times 10^6) (15 \times 10^3)}$$

$$E24 = 5,1 \text{ pF} = 5,305 \text{ pF}$$

From **Figure 2** and **Table 4**, determine V2 (nF)
2

From **Figure 2** and **Table 4**, determine V3 (pF)
2

From **Figure 2** and **Table 4**, determine V4 (nF)
2

$$L = X_{CT}$$

From **Figure 2** and **Table 4**, determine V5 (uH)
2

$$C_2 = \frac{1}{2\pi f_s R_1}$$

From **Figure 2** and **Table 4**, determine V6 (uH)
2

$$f_c = A_V \times R_1$$

From **Figure 2** and **Table 4**, determine V7 (V)
2

From **Figure 2** and **Table 4**, determine V8 (nW)
2

$$X_{CT} = \frac{1}{2\pi f C_T}$$

$$= \frac{1}{2\pi (2 \times 10^6) (100 \times 10^{-14})} \quad [16]$$

$$= 795,775$$

$$= 7,958$$

$$E24 = 8,2$$

Unit 2: [10 marks: Engineering Science]

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Table 5: Unit 2 Q

Noise Source	SPL (dB)	Sound Pressure (dyn/cm ²)
soft whisper	30	V1
rainfall	50	V2
normal conversation	60	V3
shouting in ear	110	V4
thunder	120	V5

0,63
6,32

From **Table 5**, determine V1 (Dynes per square centimeter) (milli)
2

From **Table 5**, determine V2 (Dynes per square centimeter) (milli)
2

From **Table 5**, determine V3 (Dynes per square centimeter) (milli)
2

From **Table 5**, determine V4 (Dynes per square centimeter)
2

From **Table 5**, determine V5 (Dynes per square centimeter)
2

[10]

Unit 1: [10 marks: Engineering Science]

$$V_3 = 0,0002 \times 10^{\frac{60}{20}}$$

$$= 0,02$$

$$V_4 = 0,00002 \times 10^{\frac{110}{20}}$$

$$= 6,32$$

$$V_5 = 0,00002 \times 10^{\frac{120}{20}}$$

$$= 20$$

$$V_1 = 0,00002 \times 10^{\frac{30}{20}}$$

$$= 0,63$$

$$V_2 = 0,0002 \times 10^{\frac{50}{20}}$$

$$= 6,324$$

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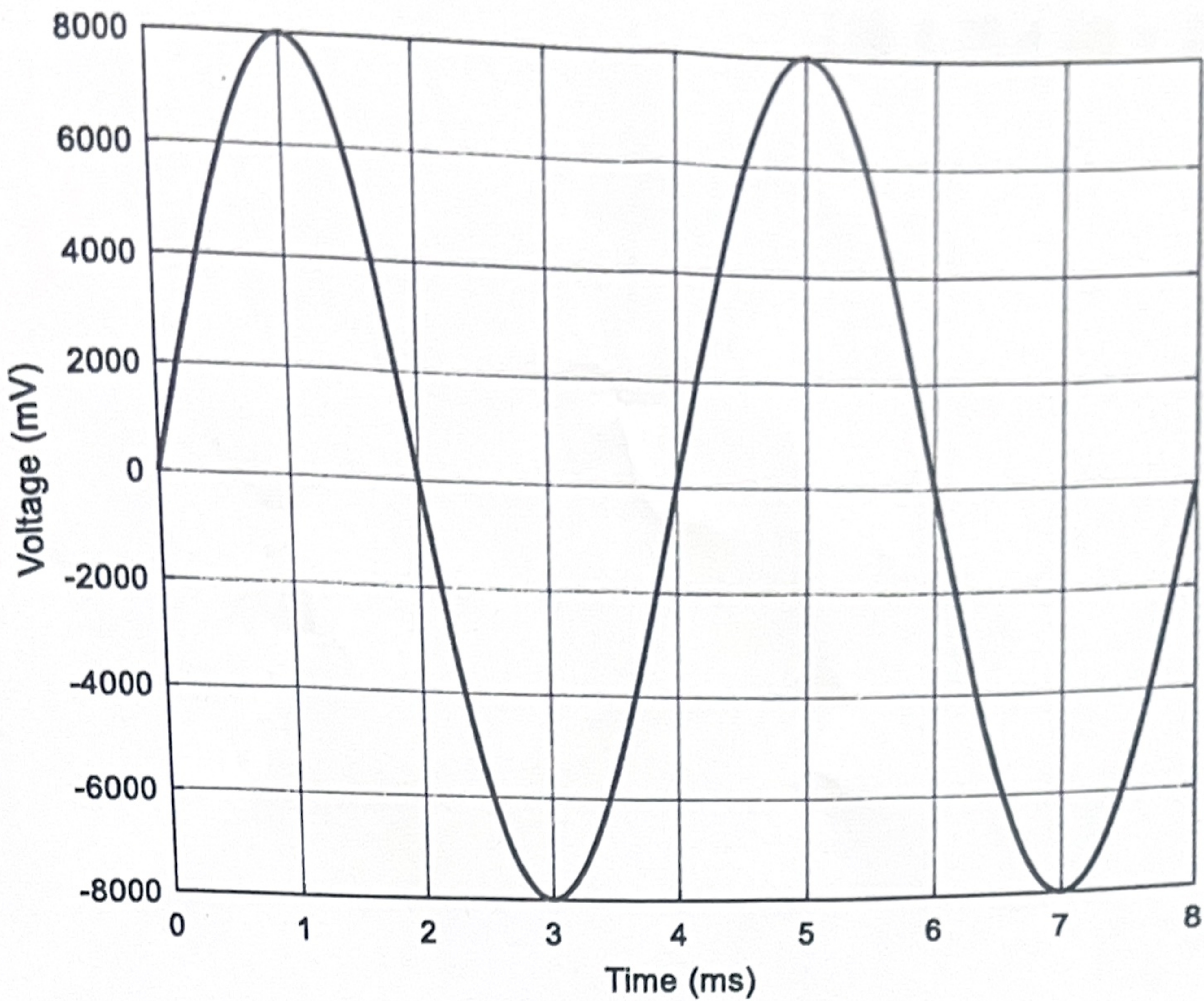


Figure 3: Unit 1 Q

Table 6: Unit 1 Q

Parameter (units)	Value
Vrms (V)	V1 2828
Vaverage (V)	V2 2548
Angular freq (rad/s) (kilo)	V3 785,398
Form factor (dimensionless)	V4 1,12
Peak factor (dimensionless)	V5 4000

$I_{rms} = 0,707 \times$

$P_{av} = V_{rms} \times I_{rms}$
 $= 11312000$

From **Figure 3** and **Table 6**, determine V1 (V)
2

$V_1 = V_p \times 0,707$
 $= 4000 \times 0,707$
 $= 2828 \checkmark$

From **Figure 3** and **Table 6**, determine V2 (V)
2

$V_2 = 4000 \times 0,637$
 $= 2548$

From **Figure 3** and **Table 6**, determine V3 (rad/s) (kilo)
2

From **Figure 3** and **Table 6**, determine V4 (dimensionless)
2

$V_4 = \frac{V_{rms}}{V_{av}} = \frac{2828}{2548} = 1,12$

From **Figure 3** and **Table 6**, determine V5 (dimensionless)
2

$\omega = 2\pi f$
 $= 2\pi(125)$
 $V_3 = 785,398$
 $f = \frac{1}{T} = \frac{1}{8} = 125 \text{ Hz}$

$V_5 = \frac{P_{av}}{V_{rms}}$
 $= \frac{11312000}{2828} = 4000$